QUICKLEAD

Digital Platform - Technical Requirements

IT Brief | Customer App + B2B Fleet Portal + Integration with Quicklead Operations

1. Purpose

Build a secure digital platform that extends the existing Quicklead operational software to customers and fleet clients. The existing operation system remains the system of record for repair orders, service status, pricing, invoicing and workshop execution.

2. Target Solution Architecture

- B2C mobile application for individual customers (iOS / Android).
- B2B web portal for fleet and corporate customers.
- API / Integration Layer between digital channels and the current Quicklead operation software.
- Central Identity & Access Management for customers, corporate users and roles.
- Notification service for push notifications, SMS and email where required.
- Payment gateway integration for B2C and future B2B payment use cases.
- Analytics and event tracking layer for customer behavior, conversion and retention.

3. Core B2C Functional Requirements

Module	Required Capability	Primary Integration
Authentication	Mobile OTP login, profile, language, consent and notification preferences	Identity / CRM
My Garage	Add multiple vehicles by plate/VIN; vehicle profile, mileage and ownership	Vehicle master
Booking	Service, branch, date/time, capacity-aware slots, reschedule/cancel	Workshop capacity / scheduling
Service History	Closed ROs, services, parts, mileage, invoices and warranty references	RO / invoice history
Live Service Status	Booked, checked-in, RO opened, in service, awaiting approval, ready, closed	RO status
Digital Approval	Estimate, photos/attachments, price, approve/reject, audit trail	RO estimate / approval
Payments & Invoices	Online payment, receipt, tax invoice and downloadable documents	Finance / payment gateway
Loyalty	Points, tiers, coupons, referral and campaign eligibility	Loyalty / CRM
Reminders	Mileage/time-based service,	CRM / notification engine

	battery, recommendations and abandoned booking	

4. B2B Fleet Portal Requirements

- Corporate account hierarchy: company, business unit, cost center, fleet, user and role.
- Vehicle master with status, service history, open repair orders and spend history.
- Dashboard for vehicles in workshop, pending approvals, completed jobs, downtime and expenditure.
- Digital approval workflow with estimate details, attachments, comments and full audit history.
- Configurable approval matrix by amount, user role, service category, vehicle or cost center.
- Commercial controls: credit limit, PO requirement, allowed services, approved branches and vehicle-specific limits.
- Finance area: invoices, credit notes, statements, outstanding balance, due dates, PDF download and Excel export.
- Reporting by vehicle, branch, service category, date period, fleet and cost center.
- Future API capability for large fleet/rental customers to exchange vehicles, approvals, ROs and invoices.

5. Integration & API Requirements

- Use API-first integration; avoid direct mobile/web access to the operational database.
- Define a canonical customer, vehicle, booking, repair order, estimate, approval and invoice model.
- Support real-time or near-real-time status synchronization for customer-facing events.
- Use idempotent APIs for booking, approval and payment actions to prevent duplicates.
- Provide webhooks/event messages for RO status changes, estimate creation, invoice issuance and booking updates.
- Maintain correlation IDs and centralized logs across app, portal, API layer and operation system.
- Version APIs and document them using OpenAPI/Swagger.

6. Security, Compliance & Controls

- Role-based access control (RBAC) and strict separation between B2C and corporate data.
- OTP/MFA where appropriate; secure token lifecycle and session management.
- Encryption in transit and at rest; secrets managed outside source code.
- Audit logs for approvals, profile changes, invoice access and administrative actions.
- Privacy-by-design: consent, data minimization, retention rules and account deletion workflow.
- Apply Saudi PDPL and applicable cybersecurity/payment requirements; IT/legal teams to validate final compliance scope.
- Regular vulnerability scanning, penetration testing and secure SDLC controls.

7. Non-Functional Requirements

- High availability for booking, approvals and service-status functions.
- Scalable architecture supporting branch expansion without redesign.
- Performance targets to be agreed for API response time, page load and notification delivery.
- Arabic and English support, including RTL/LTR behavior.
- Monitoring, alerting, error tracking, backups and disaster recovery.
- Separate development, UAT and production environments with controlled releases.

8. Recommended Delivery Phases

1. Phase 1 - MVP: authentication, vehicles, booking, branch/service selection, service history, live RO status, invoices and notifications.
2. Phase 2 - Customer Experience: loyalty, payments, digital approvals, offers, coupons, reminders, pickup/delivery options.
3. Phase 3 - B2B & Intelligence: fleet portal, approval matrix, credit/PO controls, statements, dashboards, enterprise APIs and AI-driven recommendations.

9. IT Discovery Items Before Development

- Document the current operation software database, APIs and integration constraints.
- Map every customer-facing status to an exact RO/system status.
- Confirm booking capacity logic by branch, bay, technician and service type.
- Define customer/vehicle identity matching and duplicate handling.
- Confirm payment, VAT invoice, loyalty and CRM ownership systems.
- Define B2B account hierarchy and approval rules with Operations/Finance.
- Prepare API catalogue, data dictionary, integration sequence diagrams and UAT scenarios.

10. Key Success Metrics

- Digital booking conversion and share of total appointments.
- Reduction in calls/messages asking for vehicle status.
- Approval turnaround time for B2B and additional work.
- Repeat visit rate and loyalty engagement.
- Average invoice value and accepted recommended work.
- B2B invoice/statement self-service adoption.